

Ronak Mehta.

<https://ronakrm.com>
🎓 Google Scholar  LinkedIn
ronakrm@gmail.com

Experience.

Coordinal Research. coordinal.org
Founder

San Francisco, CA
Present

- Architected production AI safety research automation framework: multi-agent orchestration with hierarchical delegation, three-tier real-time validation (local checks, policy engines, AI-judge code review), and time-travel debugging via immutable workspace snapshots.
- Built evaluation infrastructure integrating SWE-bench and RE-Bench with configurable agent scaffolding; deployed on AWS with warm EC2 instance pools, AWS Batch and IaC infrastructure for rapid benchmark iteration.
- Developed Python tooling for agent orchestration, telemetry (OpenTelemetry/Logfire), and CI/CD across specialized Docker and job requirements (GPU software stacks, $\text{\LaTeX}$, Lean4 automated theorem proving)
- Managed full organizational lifecycle: incubator/accelerator programs (Catalyze Impact, 50/50), C Corp incorporation, legal, and cloud infrastructure operations.

Machine Learning and Theory Scholars Program
Research Scholar

Berkeley, CA
June 2024 - December 2024

- Worked on theoretical and practical solutions for identifying and accounting for worst-case model behaviors.
- Built out mechanisms for finetuning efficient bounds on model performance based on model internals.
- Applied classical optimization schemes such as Lipschitz optimization and mirror descent to find jailbreaks and identify regions in the input sequence space that may exhibit outlier behaviors.

Orca DB, Inc.
Member of Technical Staff

Boston, MA
September 2023 - September 2024

- Founding scientist and engineer building out core ML business solutions and models enabling direct control and interpretability via memory inspection and editing.
- Worked on memory augmentation for machine learning models ranging from large language models to simpler classifiers and regression models for non-generative use cases.

Education.

Computer Sciences, PhD

2016 - 2022

University of Wisconsin-Madison

Thesis: Identifying Feature, Parameter, and Sample Subsets in Machine Learning and Image Analysis

Minor in Statistics

Selected Publications: ICLR 2023 (top 25%), CVPR 2022, ICCV 2019 (x3), UAI 2019, 2021 (see Google Scholar)

Computer Sciences, MS

2014 - 2016

University of Wisconsin-Madison

Computer Engineering, B.S.E.

2010 - 2014

University of Michigan-Ann Arbor

Skills.

Research Areas: Interpretability by Design, Guaranteed-Safe AI, Statistical Machine Learning, Optimal Transport, Differential Geometry, Convex/Non-convex Optimization

AI/ML Systems: LLM Agent Orchestration, Evaluation Frameworks (Inspect AI), LLM APIs, Agent Frameworks (DSPy, Pydantic-AI, Agents SDK), Coding Agents (Claude Code, Codex)

Infrastructure: Docker, IaC (Pulumi), Cloud management and job orchestration, CI/CD for full-stack AWS research automation (EC2, S3, ECR, AWS Batch, AWS VPC, CloudFront/Route53, IAM/SSM, OpenTelemetry/CloudWatch)

Languages & Web: Python (PyTorch, Transformers, sklearn, NumPy/SciPy, Pandas), TypeScript (React, Vite, Next.js, Tailwind), R (lme4, ggplot), C++, Julia